JEA (JAY) KWON

RESEARCH INTERESTS

– Machine memory and AI engrams: localizing, editing, and erasing learned knowledge in LLM parameters

– Mechanistic interpretability, steerability, and alignment of large language models

– Neuro4AI and AI4Neuro: brain-inspired AI and AI-driven neuroscience

EDUCATION

– **Ph.D. in Nano-Bio-Information-Technology**
Korea University, Seoul, South Korea 2014 - 2022
◇ Advisor: C. Justin Lee
◇ Full-Ride Scholarship, KU-KIST Graduate School

– **B.E. in Electrical and Electronic System**
Saitama University, Saitama, Japan 2008 - 2012
◇ Full-Ride Scholarship, Korea-Japan Governments Joint Scholarship Program

PROFESSIONAL EXPERIENCE

– **Postdoctoral Researcher**
Max Planck Institute for Security and Privacy (MPI-SP), Bochum, Germany 2024 - Present
◇ Advisor: Meeyoung Cha

– **Postdoctoral Researcher**
Institute for Basic Science (IBS), Daejeon, South Korea 2022 - 2024

– **Researcher**
Institute for Basic Science (IBS), Daejeon, South Korea 2018 - 2020

– **Student Researcher**
Korea Institute of Science and Technology (KIST), Seoul, South Korea 2014 - 2018

PUBLICATION SUMMARY

– **Citations**
  – citations **1006** · h-index **12** · i10-index **16** as of May 2026
  – `https://scholar.google.com/citations?user=6I0mg_EAAAAJ`

– **In Conferences**
  – *main author* (9): ICML2026, AAAI2026, ICLR2025, NeurIPS2023, ACL2026-f, ICLR2025-w, ICLR2024-w, ICLR2024-w, CVPR2023-w
  – *co-author* (8): ICML2026, ACL2026, ICLR2026, ICLR2026, ICLR2026-w, ICLR2025-b, NeurIPS2022-w, CVPR2022-w

– **In Journals**

- *main author* (6): Exp.Mol.Med.2026, IJCV2024, Exp.Neurobiol.2024, Mol.Brain2021, Exp.Neurobiol.2018, Exp.Neurobiol.2017
- *co-author* (14): Nat.Commun.2025, Nat.Commun.2024, Nat.Metab.2023, Brain2023, Mol.Brain2022, CellMetab.2022, Exp.Neurobiol.2022, Exp.Neurobiol.2021, Exp.Neurobiol.2021, J.Neurosci.2020, Curr.Biol.2019, Exp.Neurobiol.2019, PNAS2018, Nat.Commun.2016

– Suffixes: -w = workshop, -f = findings, -b = blogpost

ACHIEVEMENTS

- **Seal of Excellence — MSCA Postdoctoral Fellowships 2025** • European Commission, Horizon Europe (HORIZON-MSCA-2025-PF-01-01) • 2026 Proposal 101286042 — TRACE: Towards Reliable AI through Cognitive Engrams (with Max-Planck-Gesellschaft), total score 89.40%
- **Excellent Paper Award** • Korean Artificial Intelligence Association (JKAIA) • 2023 Brain-inspired Lp-Convolution benefits large kernels and aligns better with visual cortex
- **Patent Application & Technology Transfer** • Application Number: 10-2023-0027219 • 2023 Device and Method for Classification of Human and Animal Behavior, Patent Contribution=35% Finalized a technology transfer with ACTNOVA Inc.
- **Best Presentation Award** • Korean Society for Brain and Neural Sciences (KSBNS) • 2022 ABCD-analysis: Mapping animal behavior and differential analysis of kinematic features without selection bias
- **Young Investigator Award** • Korean Society for Molecular and Cellular Biology (KSMCB) • 2022 Retina-attached slice recording reveals light-triggered tonic GABA signaling in suprachiasmatic nucleus
- **Best Poster Award** • Center for Cognition and Sociality (CCS) Workshop • 2021 Mapping Behavior with DECLARE: Deep Embedded Clustering of Action REpresentation

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TEACHING

- **Guest Lecturer** • Georgia Tech CSE 8803 BMI "Brain-inspired Machine Intelligence", Fall 2024
- **Guest Lecturer** • Ewha Womans University G 18067 "Introduction to System Health", Spring 2026
- **Mentoring** • SoHyung Kim (Korea University, Undergraduate Intern), IBS 2020, 6 months • Outputs: CVPR2023-w, IJCV2024 (co-author)
- **Mentoring** • Hyewon Kim (Eulji University, Undergraduate Intern), IBS 2024, 9 months • Outputs: ICLR2024-w (co-author), Exp.Neurobiol.2024 (co-first author)
- **Mentoring** • Minsung Kim (Seoul National University, Ph.D. Intern, with Dong-Kyum Kim), MPI-SP 2024, 3 months • Outputs: ICLR2025-b, ACL2026 (first author)
- **Mentoring** • Jiseon Kim (KAIST, Ph.D. Intern, with Luiz Felipe Vecchietti), MPI-SP 2025, 6 months • Outputs: ICLR2025-w, ACL2026-f (first author)
- **Mentoring** • Nakyeong Yang (Seoul National University, Ph.D. Intern, with Dong-Kyum Kim), MPI-SP 2025, 3 months • Outputs: ICLR2026 (first author)

COMMUNITY SERVICE

- **Symposium Organizer** • KSBNS 2026 Annual Meeting • "Minds Meet Machines: Bidirectional Innovations in AI and Neuroscience"
- **Symposium Co-Organizer** • KSBNS & CJK 2025 Joint Symposium • "Neuroscience-inspired AI: Computational Insights into Biological and Artificial Intelligence"
- **Program Committee** • AAAI 2026, AI Alignment Track
- **Program Committee** • Machine+Behavior 2026, Harnack House, Berlin
- **Emergency Reviewer** • ICML 2026
- **Reviewer** • ICML 2026 Mechanistic Interpretability Workshop
- **Reviewer** • ICLR 2026
- **Reviewer** • ICLR 2026 Re-Align Workshop
- **Reviewer** • ICLR 2025 Re-Align Workshop
- **Founding Member** • KAIST-NeuroAI Journal Club, 2020

TALKS

- **AI Engram: In Search of Memory Traces in Artificial Neural Networks** • Scalable Trustworthy AI (STAI) group - KAIST • 2026 (Invited Seminar, In person, invited by Seong Joon Oh)
- **AI Engram: In Search of Memory Traces in Artificial Neural Networks** • Max Planck Institute for Software Systems • 2026 (Invited Seminar, In person, invited by Krishna Gummadi)
- **AI Engram: In Search of Memory Traces in Artificial Neural Networks** • Spark(l)ing Science - Max Planck Institute for Security and Privacy • 2026 (Seminar, In person)
- **How neuroscience can benefit AI** • KAIST BK21 IBS Symposium - Integrated Neuroscience and Physiology • 2025 (Invited Seminar, In person, invited by Greg Suh)
- **Transformer as a Hippocampal Memory Consolidation Model based on NMDAR-inspired Nonlinearity** • KSBNS Symposium • 2025 (Invited Seminar, In person)
- **Sovereign AI: K-AI as an opportunity** • Netzwerk Junge Generation Deutschland-Korea • 2025 (Public Lecture, In person, invited by Muhong Lee)
- **Brain-inspired AI** • Institute for Basic Science (IBS) • 2025 (Invited Seminar, In person, invited by C. Justin Lee)
- **Brain-inspired AI** • Korea Brain Research Institute (KBRI) • 2025 (Invited Seminar, In person, invited by Jeongyeon Kim)
- **Brain-inspired AI** • Gwangju Institute of Science and Technology (GIST) • 2025 (Invited Seminar, In person, invited by Sundong Kim)
- **Neuroscience for Artificial Intelligence** • MLAI group, Yonsei University • 2025 (Invited Lecture, Offline, invited by Kyungwoo Song)
- **Brain-inspired AI** • Kyungpook National University (KNU) • 2024 (Invited Seminar, In person, invited by Saekwang Nam)

- **SUBTLE: An Unsupervised Platform with Temporal Link Embedding that Maps Animal Behavior** • Korea Institute of Ocean Science & Technology (KIOST) • 2024 (Invited Seminar, In person, invited by Soomee Kim)
- **SUBTLE: An Unsupervised Platform with Temporal Link Embedding that Maps Animal Behavior** • Korea Brain Research Institute (KBRI) • 2024 (Invited Seminar, In person, invited by Jeongyeon Kim)
- **Transformer as a hippocampal memory consolidation model based on NMDAR-inspired nonlinearity** • IBS-CCS Symposium • 2023 (Oral Presentation, In person)
- **Memory editing in the brain - Eternal Sunshine of the Spotless Mind** • IBS Science in the Cinema Roof • 2023 (Public Lecture, In person)
- **From human mind to artificial intelligence - Is ChatGPT a bridge** • Reading Festival at Jeonil High School • 2023 (Public Lecture, In person)
- **Resolving clustering bias during animal behavior mapping and kinematic profiling** • IBS Data Science Group seminar • 2022 (Invited Seminar, Online, invited by Meeyoung Cha)
- **Resemblances Between Transformer's Nonlinearity and NMDA Receptor Dynamics** • IBS Winter School on AI-Boosted Basic Science • 2022 (Invited Lecture, In person)
- **Age-dependent motor coordination analysis by 3D pose-estimation using AVATAR** • IBS 2021 2nd Workshop • 2021 (Oral Presentation, In person)
- **Backpropagation and the Brain** • KAIST-NeuroAI Journal Club • 2020 (Oral Presentation, Hybrid)
- **Deep learning with NMDA receptor inspired activation function** • IBS 2019 Year-End Workshop • 2020 (Oral Presentation, In person)
- **Development of a Low-cost, Comprehensive Recording System for Circadian Rhythm Behavior** • Korean Society for Molecular and Cellular Biology (KSMCB) • 2019 (Oral Presentation, In person)

PUBLICATIONS

✓ selected publication * first author # corresponding author

✓ **J. Kwon***, D. Kim*, J. Kim*, Y. Kim, W. Kook, M. Cha. AI Engram: In Search of Memory Traces in Artificial Intelligence. *International Conference on Machine Learning*, 2026 (Spotlight, top 2.2%).

M. S. Locatelli, F. Tonucci, **J. Kwon**, L. F. Vecchietti, B. N. Wijaya, C. Y. Low, V. Almeida, M. Cha. Textual Supervision Enhances Geospatial Representations in Vision-Language Models. *International Conference on Machine Learning*, 2026 (Acceptance Rate=26.6%).

✓ J. Kim, **J. Kwon#**, L. F. Vecchietti#, W. Dong, J. Kim, M. Cha#. Machine Behavior in Relational Moral Dilemmas: Moral Rightness, Predicted Human Behavior, and Model Decisions. *Findings of the 64th Annual Meeting of the Association for Computational Linguistics*, 2026 (Acceptance Rate=18%).

M. Kim, D. Kim, **J. Kwon**, N. Yang, K. Jung, M. Cha. How Training Data Shapes the Use of Parametric and In-Context Knowledge in Language Models. *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*, 2026 (Acceptance Rate=19%).

S. Kim, **J. Kwon**. From Human Cognition to AI Reasoning: Models, Methods, and Applications. *ICLR HCAIR Workshop*, 2026.

✓ **J. Kwon***, S. Kim*, J. Woo, K. Tanaka-Yamamoto, O. James, E. De Schutter, S. Hong, C. J. Lee. Cerebellar Tonic Inhibition Orchestrates the Maturation of Information Processing and Motor Coordination. *Experimental & Molecular Medicine*, 2026 (5-year IF=14.2, 2024).

✓ **J. Kwon***, L. F. Vecchietti, S. Park, M. Cha. Dropouts in Confidence: Moral Uncertainty in Human-LLM Alignment. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026 (AI Alignment Track, Acceptance Rate=22.65%).

D. Kim, M. Kim, **J. Kwon**, N. Yang, M. Cha. Bilinear Relational Structure Fixes Reversal Curse and Enables Consistent Model Editing. *International Conference on Learning Representations*, 2026 (Acceptance Rate=28%).

N. Yang, D. Kim, **J. Kwon**, M. Kim, K. Jung, M. Cha. Erase or Hide? Suppressing Spurious Unlearning Neurons for Robust Unlearning. *International Conference on Learning Representations*, 2026 (Acceptance Rate=28%).

M. Kim, **J. Kwon**, D. Kim, M. Cha. In Search of the Engram in LLMs: A Neuroscience Perspective on the Memory Functions in AI Models. *ICLR Blogposts Track*, 2025.

B. Hyeon, J. Shin, J. Lee, W. Kim, **J. Kwon**, H. Lee, D. Kim, C. Y. Kim, S. Choi, J. Jeong, K. Kim, C. J. Lee, D. Kim, W. D. Heo. Integrating Artificial Intelligence and Optogenetics for Parkinson's Disease Diagnosis and Therapeutics in Male Mice. *Nature Communications*, 2025.

J. Kim*, **J. Kwon***, L. F. Vecchietti*, A. Oh, M. Cha. Exploring Persona-dependent LLM Alignment for the Moral Machine Experiment. *ICLR Bi-Align Workshop*, 2025.

✓ **J. Kwon***, S. Lim, K. Song, C. J. Lee. Brain-inspired Lp-Convolution Benefits Large Kernels and Aligns Better with Visual Cortex. *International Conference on Learning Representations*, 2025 (Acceptance Rate=32.08%).

J. Kwon*, M. Sa, H. Kim, Y. Seong, C. J. Lee. Egocentric 3D Skeleton Learning in Identity-Aware Deep LSTM Network Encodes Obese-Like Motion Representations. *ICLR TS4H Workshop*, 2024.

J. Kwon*, S. Lim, K. Song, C. J. Lee. Brain-inspired Lp-Convolution Benefits Large Kernels and Aligns Better with Visual Cortex. *ICLR Re-Align Workshop*, 2024.

✓ **J. Kwon***, S. Kim, D. Kim, J. Joo, S. Kim, M. Cha, C. J. Lee. SUBTLE: An Unsupervised Platform with Temporal Link Embedding that Maps Animal Behavior. *International Journal of Computer Vision*, 2024 (5-year IF=15.5, 2024).

✓ **J. Kwon***, M. Sa, H. Kim, Y. Seong, C. J. Lee. Egocentric 3D Skeleton Learning in a Deep Neural Network Encodes Obese-like Motion Representations. *Experimental Neurobiology*, 2024 (Featured on the cover).

H. Kang, A. Han, A. Zhang, H. Jeong, W. Koh, J. M. Lee, H. Lee, H. Y. Jo, M. A. Maria-Solano, M. Bhalla, **J. Kwon**, et al.. GolpHCat (TMEM87A), a Unique Voltage-Dependent Cation Channel in Golgi Apparatus, Contributes to Golgi-pH Maintenance and Hippocampus-Dependent Memory. *Nature Communications*, 2024.

✓ D. Kim*, **J. Kwon***, M. Cha, C. J. Lee. Transformer as a Hippocampal Memory Consolidation Model Based on NMDAR-Inspired Nonlinearity. *Advances in Neural Information Processing Systems*, 2023 (Acceptance Rate=26.1%).

M. Sa, E. Yoo, W. Koh, M. G. Park, H. Jang, Y. R. Yang, M. Bhalla, J. Lee, J. Lim, W. Won, **J. Kwon**, et al.. Hypothalamic GABRA5-Positive Neurons Control Obesity via Astrocytic GABA. *Nature Metabolism*, 2023.

J. Kwon*, S. Kim, D. Kim, J. Joo, S. Kim, M. Cha, C. J. Lee. SUBTLE: An Unsupervised Platform with Temporal Link Embedding that Maps Animal Behavior. *CVPR CV4Animals Workshop*, 2023.

M. Nam, H. Y. Ko, D. Kim, S. Lee, Y. M. Park, S. J. Hyeon, W. Won, J. Chung, S. Y. Kim, H. H. Jo, K. T. Oh, Y. Han, G. Lee, Y. H. Ju, H. Lee, H. Kim, J. Heo, M. Bhalla, K. J. Kim, **J. Kwon**, et al.. Visualizing Reactive Astrocyte-Neuron Interaction in Alzheimer's Disease Using 11C-Acetate and 18F-FDG. *Brain*, 2023.

D. Kim, **J. Kwon**, M. Cha, C. J. Lee. Transformer Needs NMDA Receptor Nonlinearity for Long-Term Memory. *NeurIPS MemARI Workshop*, 2022.

D. Kim, J. Kim, W. Jung, J. Park, M. Kim, A. Shin, Y. Jeong, S. Park, G. Shin, Y. W. Lee, **J. Kwon**, D. Kim. AVATAR: AI Vision Analysis for Three-Dimensional Action in Real-Time. *CVPR CV4Animals Workshop*, 2022.

S. Kim, **J. Kwon**, M. G. Park, C. J. Lee. Dopamine-Induced Astrocytic Ca2+ Signaling in mPFC is Mediated by MAO-B in Young Mice, but by Dopamine Receptors in Adult Mice. *Molecular Brain*, 2022.

Y. H. Ju, M. Bhalla, S. J. Hyeon, J. E. Oh, S. Yoo, U. Chae, **J. Kwon**, W. Koh, J. Lim, Y. M. Park, et al.. Astrocytic Urea Cycle Detoxifies Aβ-Derived Ammonia While Impairing Memory in Alzheimer's Disease. *Cell Metabolism*, 2022.

J. M. Lee, M. Sa, H. An, J. M. J. Kim, **J. Kwon**, B. Yoon, C. J. Lee. Generation of Astrocyte-Specific MAOB Conditional Knockout Mouse with Minimal Tonic GABA Inhibition. *Experimental Neurobiology*, 2022.

✓ **J. Kwon***, M. W. Jang, C. J. Lee. Retina-Attached Slice Recording Reveals Light-Triggered Tonic GABA Signaling in Suprachiasmatic Nucleus. *Molecular Brain*, 2021.

M. W. Jang, T. Y. Kim, K. Sharma, **J. Kwon**, E. Yi, C. J. Lee. A Deafness Associated Protein TMEM43 Interacts with KCNK3 (TASK-1) Two-Pore Domain K+ (K2P) Channel in the Cochlea. *Experimental Neurobiology*, 2021.

J. Won, H. H. Kazan, **J. Kwon**, M. Park, M. A. Ergun, S. Ozcan, B. Y. Choi, W. D. Heo, C. J. Lee. Ultimate COVID-19 Detection Protocol Based on Saliva Sampling and qRT-PCR with Risk Probability Assessment. *Experimental Neurobiology*, 2021.

K. Han, M. Lee, H. Lim, M. W. Jang, **J. Kwon**, C. J. Lee, S. Kim, M. Suh. Excitation-Inhibition Imbalance Leads to Alteration of Neuronal Coherence and Neurovascular Coupling Under Acute Stress. *Journal of Neuroscience*, 2020.

S. Oh, J. M. Lee, H. Kim, J. Lee, S. Han, J. Y. Bae, G. Hong, W. Koh, **J. Kwon**, E. Hwang, et al.. Ultrasonic Neuromodulation via Astrocytic TRPA1. *Current Biology*, 2019.

Y. Han, **J. Kwon**, J. Won, H. An, M. W. Jang, J. Woo, J. S. Lee, M. G. Park, B. Yoon, S. E. Lee, et al.. Tweety-Homolog (Ttyh) Family Encodes the Pore-Forming Subunits of the Swelling-Dependent Volume-Regulated Anion Channel (VRACswell) in the Brain. *Experimental Neurobiology*, 2019.

J. Woo, J. O. Min, D. Kang, Y. S. Kim, G. H. Jung, H. J. Park, S. Kim, H. An, **J. Kwon**, J. Kim, et al.. Control of Motor Coordination by Astrocytic Tonic GABA Release Through Modulation of Excitation/Inhibition Balance in Cerebellum. *Proceedings of the National Academy of Sciences*, 2018.

J. Kwon*, M. G. Park, S. E. Lee, C. J. Lee. Development of a Low-Cost, Comprehensive Recording System for Circadian Rhythm Behavior. *Experimental Neurobiology*, 2018.

J. Kwon*, H. An, M. Sa, J. Won, J. I. Shin, C. J. Lee. Orai1 and Orai3 in Combination with Stim1 Mediate the Majority of Store-Operated Calcium Entry in Astrocytes. *Experimental Neurobiology*, 2017.

G. E. Ha, J. Lee, H. Kwak, K. Song, **J. Kwon**, S. Jung, J. Hong, G. Chang, E. M. Hwang, H. Shin, et al.. The Ca2+-Activated Chloride Channel Anoctamin-2 Mediates Spike-Frequency Adaptation and Regulates Sensory Transmission in Thalamocortical Neurons. *Nature Communications*, 2016.